Correlation-Aware Selective Reward Acquisition for Group-Relative Policy Updates

Shenggang Li[1] Bilian Song[1] Yunfeng Qi[1] Joseph Li[2]

[1]Datatist, 12F Yuyan Building, 738 Dongfang Road, Pudong, Shanghai, China

[2]University of Waterloo, Waterloo, Canada

kelvin.li@datatist.com, sophia@datatist.com,

frankie@datatist.com, j3299li@uwaterloo.ca

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Abstract

Evaluating every response in a group-relative policy update may be expensive. We study how to select exactly B of G candidate rewards when prediction errors are correlated. For an unnormalized leave-one-out (LOO) gradient, a model-assisted Horvitz–Thompson estimator is design-unbiased. Its additional query risk depends jointly on residual second moments, policy-score geometry, and the dependence among selection indicators. Independent Bernoulli queries cannot exploit off-diagonal residual moments. At fixed marginal probabilities, the optimal fixed-size coupling is a finite linear program, equivalent to classical auxiliary balancing in a lifted feature space when the residual second-moment matrix is positive semidefinite. Synthetic experiments show gains under some signed or changing correlations, but not across all structures. In three-seed Qwen2.5-0.5B LoRA training, budgeted joint and uniform acquisition yield identical final held-out accuracy. A subsequent 144-group frozen-checkpoint diagnostic finds a 26.91% pooled reduction in sketch risk but only 3.27% in actual LoRA-gradient risk after warmup; one of three seeds has higher true risk. The analysis concerns a linear LOO estimator, not normalized, clipped GRPO or superior downstream accuracy.

Keywords: group-relative policy gradients; selective reward observation; dependent sampling; Horvitz–Thompson estimation; residual covariance; balanced sampling.

1 Introduction

A language model can generate several plausible responses to one prompt, although reliable rewards may be available for only a subset. A reward may require an executable test, a separate verifier, or human judgment. Suppose a group contains six responses and the evaluation budget covers three. Uniform sampling is a natural starting point, but it ignores how the responses are related. In a group-relative update, the reward assigned to one response also affects the comparison against the others. The three responses selected for evaluation therefore matter as a *set*, not merely as three separate choices.

Reward predictors offer a way to use information from all six candidates. Before making any factual queries, a model can supply a provisional reward for each response. We can then correct the predictions for the responses that are evaluated, using their known inclusion probabilities. This correction preserves a full-feedback target in expectation. It leaves a second question open: among the many ways to choose three responses, which produces the smallest additional gradient error?

The answer depends on more than the uncertainty attached to each predicted reward. Responses to the same prompt may share sources of prediction error. Their policy-score vectors may also point in different directions, so that errors can reinforce or cancel in a gradient estimate. Two acquisition designs with the same probability of selecting each response can consequently have different risks. To use this information, selection indicators must be sampled jointly. Simply inserting a full residual covariance matrix into an independent Bernoulli sampling rule does not exploit its off-diagonal entries.

We make this relationship explicit for an unnormalized leave-one-out (LOO) score-function gradient. A model-assisted Horvitz–Thompson correction gives a design-unbiased estimator, and its conditional mean-squared error separates the contribution of reward-prediction residuals, the geometry of the policy scores, and the dependence imposed by the query design. With a fixed number of queries, optimizing that dependence becomes a finite linear program. The formulation is closely related to classical balanced sampling: for positive-semidefinite residual moments, its objective is exactly a lifted auxiliary-variable imbalance criterion. We treat this relationship as a limitation on claims of new sampling theory, not as a distinction to conceal.

We test the acquisition criterion first under controlled residual structures, then with moments learned from historical observations, and finally with a small language model. Synthetic tests identify settings in which joint selection lowers conditional gradient risk and others in which it does not. Actual LoRA training yields no final held-out accuracy advantage over uniform acquisition in three completed seed panels. A separate frozen-checkpoint diagnostic then evaluates the same proposed query designs with genuine per-completion LoRA autograd, exposing a measurable discrepancy between the sketch objective and parameter-gradient risk. These experiments distinguish acquisition-risk control from its consequences for model training.

2 Related work and scope

Group Relative Policy Optimization (GRPO) was introduced for mathematical reasoning in Shao et al. [5]. Deployed GRPO variants commonly standardize group rewards and use a clipped token-level objective. We isolate a linear LOO estimator because its missing-reward properties admit exact design-based statements. The proofs below do not establish unbiasedness for the nonlinear standardized objective.

The inverse-probability step is the Horvitz–Thompson principle [3], and pre-query prediction is a standard model-assisted control variate. Balanced sampling and its cube method already seek samples that match known auxiliary totals [2]. Determinantal designs offer another way to build dependent inclusion indicators [4]. Optimal survey schemes have also been analyzed for stochastic-gradient computation [1]. We therefore do not attribute the generic estimator, LP formulation, square-root marginal rule, or balancing identity to a new sampling theory.

Related reward-cost work tackles different interventions. Su et al. [7] adapt surrogate reward models over policy learning, while Shen et al. [6] choose informative preference labels for reward-model fitting. Here the target is the variance of a *given* group-relative policy update, and the acquisition decision buys candidate-level factual rewards under known sampling probabilities.

3 Estimand and pre-query information

Let X be a context, $A \sim \pi_\theta(\cdot \mid X)$ an action, and $\mu(x, a) = \mathbb{E}[R \mid X = x, A = a]$. The one-step objective is

$$J(\theta) = \mathbb{E}_X \mathbb{E}_{A \sim \pi_\theta(\cdot|X)}[\mu(X, A)]. \tag{1}$$

Conditional on X, generate $G \geq 2$ independent candidates $A_1, \ldots, A_G$ from the current policy, and define their score vectors $s_i = \nabla_\theta \log \pi_\theta(A_i \mid X)$. Put $S = [s_1, \ldots, s_G] \in \mathbb{R}^{d \times G}$ and

$$H = I_G - \frac{\mathbf{1}\mathbf{1}^\top}{G}, \qquad L = \frac{G}{G-1}H, \qquad M = \frac{1}{G}SL = \frac{1}{G-1}SH. \tag{2}$$

The full-feedback LOO gradient is $g_{\text{full}} = MR$ for $R = (R_1, \ldots, R_G)^\top$. Its ith advantage is $R_i - (G-1)^{-1}\sum_{j \neq i} R_j$.

Let $\mathcal{F}_0$ contain the historical log, current context, candidate responses, score vectors, model predictions, and available costs, but exclude the current hidden factual rewards. Write

$$\begin{aligned} &\bar{\mu} = \mathbb{E}[R \mid \mathcal{F}_0], \quad \Sigma_R = \mathrm{Cov}(R \mid \mathcal{F}_0), \quad m \in \mathbb{R}^G \ \mathcal{F}_0\text{-measurable}, \\ &V = \mathbb{E}[(R-m)(R-m)^\top \mid \mathcal{F}_0] \\ &\quad = \Sigma_R + (\bar{\mu} - m)(\bar{\mu} - m)^\top. \end{aligned} \tag{3}$$

In particular, a model posterior covariance need not equal V when the prediction is biased.

The query vector $Z \in \{0,1\}^G$ is drawn from $Q(\cdot \mid \mathcal{F}_0)$. Let $\pi_i = \mathbb{E}[Z_i \mid \mathcal{F}_0] > 0$, $\pi_{ij} = \mathbb{E}[Z_i Z_j \mid \mathcal{F}_0]$, and $\pi_{ii} = \pi_i$. We require $Z \perp R \mid \mathcal{F}_0$; the probabilities refer to the design *actually sampled*, including any exploration mixture. The main finite-budget designs satisfy $\sum_i Z_i = B$ almost surely.

Theorem 1 (LOO gradient target)**.** *Assume the candidates are conditionally independent and identically distributed, $\mathbb{E}[R_i \mid X, A_{1:G}] = \mu(X, A_i)$, differentiation may pass through the expectation in* (1)*, and the relevant moments are finite. Then $\mathbb{E}[g_{\text{full}}] = \nabla_\theta J(\theta)$.*

The condition on the reward mean permits dependent zero-mean reward noise across candidates, but excludes mean-level interference from the other generated responses. A proof is in Appendix A.

4 Risk from selective reward observation

Define the model-assisted reward estimator and its gradient:

$$\widehat{R}_i = m_i + \frac{Z_i}{\pi_i}(R_i - m_i), \qquad \widehat{g} = M\widehat{R}. \tag{4}$$

Only queried R_i are evaluated. The unqueried entries in (4) are computed from m_i.

Theorem 2 (Design unbiasedness and covariance)**.** *Under pre-query independence and $\pi_i > 0$, for every fixed $(R, \mathcal{F}_0)$,*

$$\mathbb{E}_Z[\widehat{R} \mid R, \mathcal{F}_0] = R, \qquad \mathbb{E}_Z[\widehat{g} \mid R, \mathcal{F}_0] = g_{\text{full}}. \tag{5}$$

Let C_Q have entries $(C_Q)_{ij} = \pi_{ij}/(\pi_i \pi_j) - 1$. Then

$$\mathrm{Cov}(\widehat{R} \mid \mathcal{F}_0) = \Sigma_R + C_Q \odot V, \qquad \mathrm{Cov}(\widehat{g} \mid \mathcal{F}_0) = M(\Sigma_R + C_Q \odot V)M^\top. \tag{6}$$

The conditional mean-squared error due to querying alone is

$$\mathcal{R}(Q; V, M) = \mathbb{E}[\|\widehat{g} - g_{\text{full}}\|_2^2 \mid \mathcal{F}_0] = \sum_{i,j=1}^{G} (C_Q)_{ij} V_{ij} K_{ij}, \qquad K = M^\top M. \tag{7}$$

Equation (7) identifies the role of each source of dependence. The entry V_{ij} describes the joint prediction residuals, including any systematic prediction bias. The entry K_{ij} is the inner product of the two candidates' gradient contributions. The entry $(C_Q)_{ij}$ measures whether the two candidates are queried together more or less often than independent selection would imply. Their product determines the pair's contribution to the additional gradient error. Correlated residuals need not favor any particular pairing without the corresponding policy-score geometry.

The first term in (6), $M\Sigma_R M^\top$, is reward uncertainty that remains even with complete feedback; the second is uncertainty due to querying only a subset. Design-unbiased gradient estimation requires positive *individual* inclusion probabilities. Estimating residual cross-products directly from jointly purchased rewards also requires $\pi_{ij} > 0$. At $B = 1$, such pairwise moments cannot be learned without additional assumptions.

Proposition 3 (Bernoulli factorization)**.** *If Z_i are conditionally independent Bernoulli draws, then*

$$\mathcal{R}(Q; V, M) = \sum_i \Big(\frac{1}{\pi_i} - 1\Big) V_{ii} \|M_{\cdot i}\|_2^2. \tag{8}$$

Thus varying off-diagonal entries of V while fixing its diagonal cannot change the variance-minimizing Bernoulli probabilities. Under an interior expected-cost constraint $\sum_i c_i \pi_i = C$, they obey $\pi_i \propto \sqrt{V_{ii}\|M_{\cdot i}\|_2^2/c_i}$, with floors and upper caps applied through a multiplier.

Under independent selection, the off-diagonal entries of C_Q vanish. Residual variances can influence the individual inclusion probabilities through the diagonal of V, but off-diagonal residual moments cannot affect (8). Exploiting those moments requires dependent selection. The Bernoulli result assumes an *expected* query count; selecting exactly B candidates already induces dependence except in degenerate cases.

4.1 Fixed-size coupling and its classical reduction

For fixed feasible marginals $\pi_i > 0$ with $\sum_i \pi_i = B$, let $\mathcal{T}_B$ denote all B-subsets and $q_T = Q(Z = \mathbf{1}_T \mid \mathcal{F}_0)$. Minimizing (7) over designs with those marginals gives

$$\begin{aligned} \min_{q \geq 0} \quad & \sum_{T \in \mathcal{T}_B} q_T \sum_{i,j \in T} \frac{V_{ij} K_{ij}}{\pi_i \pi_j}, \\ \text{subject to} \quad & \sum_T q_T = 1, \qquad \sum_{T \ni i} q_T = \pi_i \quad (i = 1, \ldots, G). \end{aligned} \tag{9}$$

The omitted term $-\sum_{ij} V_{ij} K_{ij}$ is constant over q. Equation (9) is an ordinary finite linear program with $\binom{G}{B}$ columns; its application-specific coefficients combine reward residuals and policy scores. An approximate subset-pool solver is not an exact oracle.

Proposition 4 (Boundaries and a balancing identity)**.** *(i) At $B = 1$ or $B = G - 1$, fixed feasible marginals uniquely specify q; there is no coupling freedom. If $V_{ij}K_{ij}/(\pi_i\pi_j) = u_i + u_j$ for all $i \neq j$, the pairwise part of the risk is the same for every exact-B design with those marginals.*

(ii) For every $V \succeq 0$ choose vectors t_i such that $V_{ij} = t_i^\top t_j$, and set $w_i = t_i \otimes M_{\cdot i}$. Then

$$\mathcal{R}(Q; V, M) = \mathbb{E}_Z \left[\left\| \sum_i \left(\frac{Z_i}{\pi_i} - 1 \right) w_i \right\|_2^2 \middle| \mathcal{F}_0 \right]. \tag{10}$$

Hence the PSD plug-in coupling objective is a classical expected auxiliary-imbalance objective after a tensor-product feature lift.

The population matrix V in (3) is positive semidefinite by construction. An estimated plug-in matrix $\widehat{V}$ may require a PSD-preserving construction or correction for the lifted interpretation to apply. This identity makes the novelty boundary precise. The optimization does not define an entirely new class of sampling algorithms. Its relevance here comes from which lifted features are appropriate for a group-relative gradient, how V can be estimated with limited reward queries, and whether the theoretical risk predicts useful downstream behavior.

For intuition, take $G = 4$, $B = 2$, $M = (-1, -1, 1, 1)$ and $\pi_i = 1/2$, with common residual $R_i - m_i = U$, $\mathbb{E}[U^2] = 1$. Selecting $\{1, 3\}$ or $\{2, 4\}$ with probability $1/2$ has zero query risk. Selecting $\{1, 2\}$ or $\{3, 4\}$ with the same probabilities has risk 16. Both designs have identical first-order inclusion probabilities. The grouping of queried responses, together with score geometry, determines whether the shared residual cancels.

4.2 What calibration error permits

Let $W_Q = C_Q \odot K$, so $\mathcal{R}(Q; V, M) = \langle W_Q, V \rangle_{\mathrm{F}}$. If $Q^\star$ minimizes risk over a fixed design class $\mathcal{Q}$, and $\widehat{Q}$ minimizes the same criterion with $\widehat{V}$, then

$$0 \le \mathcal{R}(\widehat{Q}; V, M) - \mathcal{R}(Q^\star; V, M) \le 2 \sup_{Q \in \mathcal{Q}} \|W_Q\|_{\mathrm{F}} \, \|\widehat{V} - V\|_{\mathrm{F}}. \tag{11}$$

This deterministic comparison uses no distributional approximation; it may be uninformative when small inclusion probabilities make the supremum large. It also presumes a common feasible class, rather than different floors or marginals for the two optimizers.

5 Implementation and evaluation protocol

The pipeline uses a pre-query predictor m and a historically estimated residual second moment $\widehat{V}$. Positive marginal probabilities are assigned separately from the exact-B coupling. The design is computed using only historical observations and current candidate features, then sampled; factual rewards of unselected responses are hidden from the update. A randomized component maintains individual support and helps calibrate later moments. Pairwise moment estimation needs its own pair-support checks. A posterior covariance is not substituted for (3) without a residual-bias correction.

We first compare the designs under controlled residual structures, where their conditional gradient risks can be evaluated directly. Subsequent synthetic tests estimate residual moments from historical queries and assess the selected designs under changing conditions. The model experiments examine LoRA training and, separately, conditional risk at frozen LoRA checkpoints using per-completion autograd. All reported intervals summarize seed-level contrasts descriptively: parameter grids, models, and calibration choices were explored during development, without external preregistration. Repeated prompts or evaluation checkpoints are not independent experimental replications.

6 Experiments and results

6.1 Controlled correlated designs

The executed small-group study contains 1,008 synthetic design cells, of which 864 have interior marginals and 144 cover boundary cases. It compares uniform sampling, a cube-style balance heuristic, a globally optimized diagonal-scaled balancing baseline, and a joint LP using a pilot estimate of V. All LP coupling comparisons at fixed marginals use the same reward-query budget. In the designated

Table 1: Controlled acquisition-risk reduction, diagonal-scaled balancing risk minus estimated-joint risk. Positive entries favor joint coupling. Each row uses 36 matched interior cells, 12 seeds. All quantities are synthetic, not policy return.

Residual structure	Mean reduction	Descriptive 95% seed interval
Independent	0.00000	numerical zero
Weak positive factor	−0.00446	[−0.00947, 0.00055]
Strong positive factor	−0.00185	[−0.00503, 0.00133]
Signed factor	0.60262	[0.35773, 0.84751]
Negative equicorrelation	0.22715	[0.15233, 0.30197]
Two blocks	0.21734	[0.06305, 0.37163]

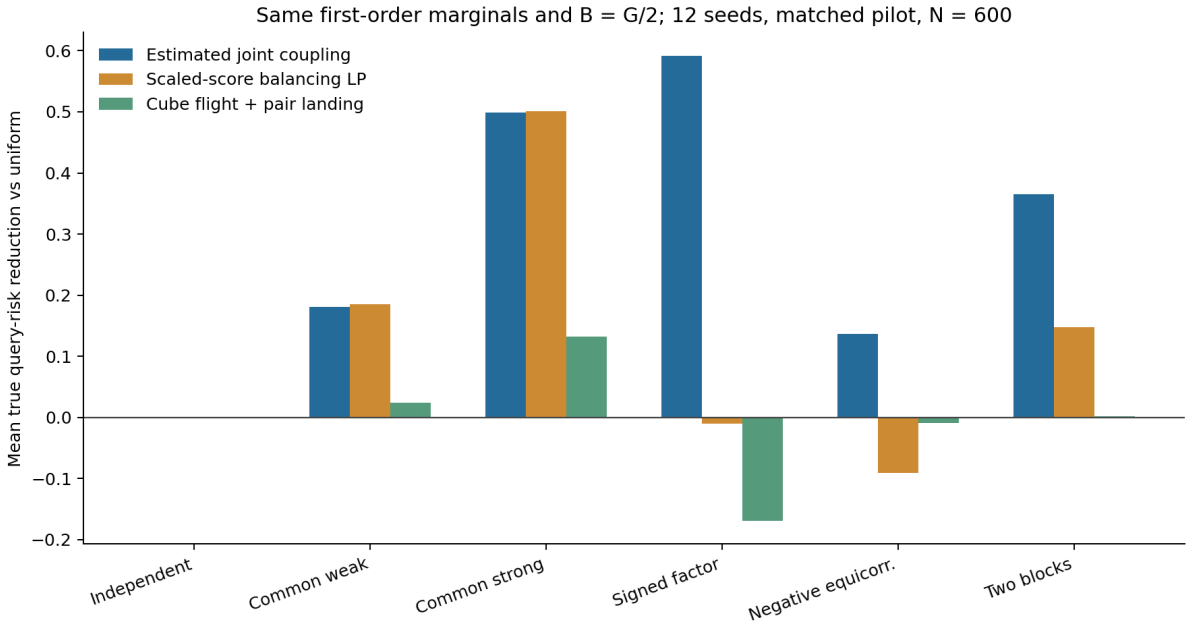


Figure 1: Synthetic true query-risk reduction relative to uniform acquisition by residual profile. Estimated joint coupling, diagonal-scaled balancing LP, and the cube-style heuristic share first-order marginals. Chart reproduced from the archived 12-seed matched-pilot study.

exploratory matched-pilot interior subset (216 cells; 12 independent seeds), mean risk reduction of joint relative to diagonal-scaled balance is 0.17347 squared-gradient units, with descriptive seed-cluster t interval $[0.12716, 0.21977]$.

The matched comparisons show no gain under independent residuals, to floating-point precision. Diagonal-scaled balancing performs at least as well for positive common factors, while the pilot-joint LP performs better for signed and block residuals (Table 1). These results also explain the need for a strong balancing comparator: improvement over uniform acquisition alone would not isolate the value of estimating residual dependence.

6.2 Historical calibration and distribution changes

An exploratory online study uses 8 seeds, five covariance-change scenarios and 38,400 design-time observations. A later internally frozen synthetic holdout comprises seven new worlds, 12 new seeds and 94,080 rows; its parameters were locked before that holdout but were not externally preregistered.

Table 2: Later conditional gradient risk in six synthetic worlds. Lower is better. The final column is scaled minus joint EWMA; negative values are losses.

World	Scaled	Joint EWMA	Difference
Independent (stationary)	0.59326	0.59326	0.00000
Positive factor (stationary)	0.47447	0.53202	−0.05755
Sparse heavy-tail (stationary)	0.59369	0.58004	0.01366
Abrupt sign switch	0.63216	0.48780	0.14437
Two block changes	0.56895	0.51584	0.05311
Smooth rotation	0.51520	0.54945	−0.03425

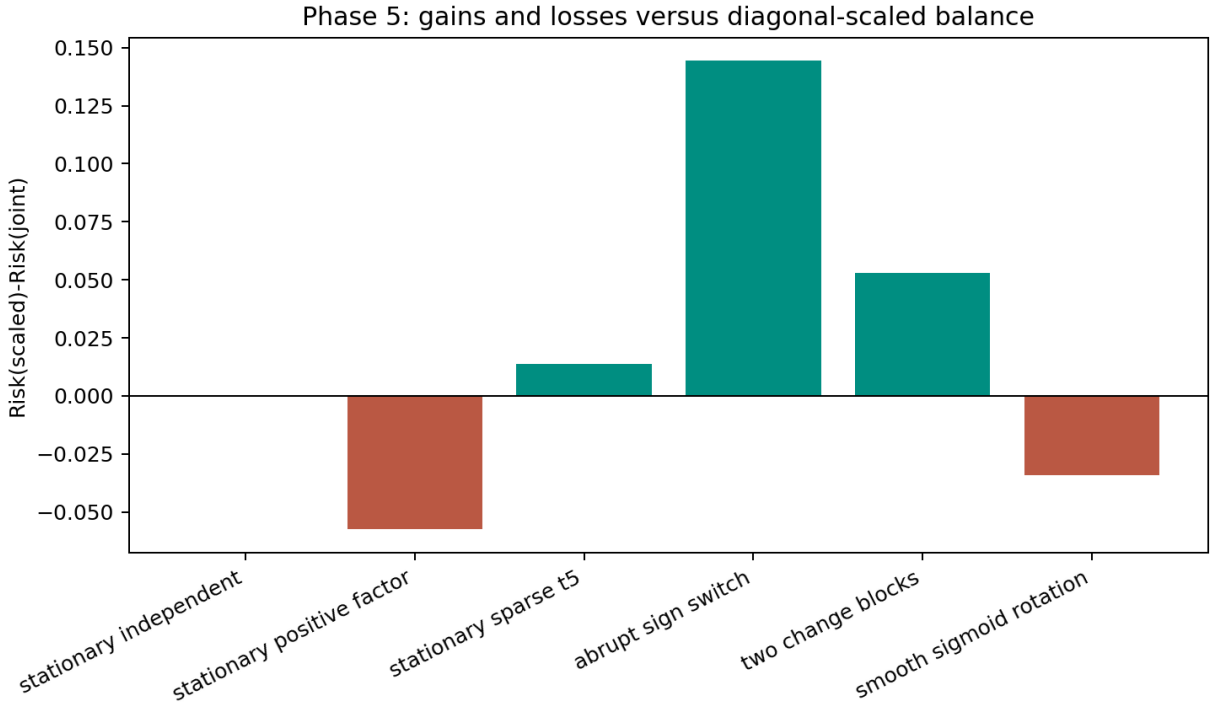


Figure 2: Paired synthetic risk reductions of adaptive joint coupling over diagonal-scaled balancing across six test worlds. The positive and negative bars correspond to the exact values in Table 2.

The joint-EWMA method lowers later conditional risk relative to uniform in all seven holdout worlds (paired mean reductions range from 0.0877 to 0.1620). This evaluation uses oracle moments *only for scoring the selected designs*; it makes no claim about realized model training accuracy.

To inspect failure modes with a more demanding comparator, a further six-world study uses 16 new seeds and 76,800 synthetic method-step rows (230, 400 metered synthetic gold queries). The average later conditional risks are 0.61760 (uniform), 0.58886 (cube-style), 0.56296 (diagonal-scaled), 0.54307 (joint EWMA), and 0.59092 (historically frozen joint design). The seed-paired mean reduction of joint EWMA relative to scaled balance is 0.01989 with descriptive interval $[0.01562, 0.02416]$; the model and scenario grid had been selected during development.

As Table 2 shows, joint EWMA outperforms scaled balancing in three worlds, ties under independence, and loses under a stationary positive factor and smooth rotation. The pooled improvement therefore depends on the mixture of residual structures. These synthetic results do not establish a uniform advantage over classical balancing.

Table 3: Deterministic greedy held-out accuracy, 24 prompts per seed. Each seed uses a shared panel across arms, but panels differ across seeds. Full feedback consumes twice the factual labels of either budgeted arm.

Seed	Initial (all arms)	Full, final	Uniform, final	Joint EWMA, final
6207	18/24	23/24	23/24	23/24
6219	17/24	24/24	24/24	24/24
6251	14/24	18/24	17/24	17/24
Pooled across seed panels	49/72	65/72	64/72	64/72

6.3 Calibrated Qwen LoRA policy updates

For the model test, a pilot study compared multiple arithmetic task families using Qwen2.5-0.5B-Instruct. Two-digit addition was chosen after exploratory calibration, because its groups offered multiple positive and negative candidate rewards. In the three-seed calibration evaluation, 67.78% of groups had multiple positive outcomes. The oracle was local exact arithmetic checking: reward counts are *logical label budgets*, with no meaningful wall-clock verifier cost. The task family choice is post-pilot and cannot be described as preregistered task selection.

The frozen Phase 6C configuration then runs 60 training steps per arm at each of seeds 6207, 6219 and 6251, with $G = 6$, $B = 3$, warmup 12, and a 9-group moment window. All three arms use the same 0.5B base model and rank-8 LoRA architecture, learning rate 10^{-5}, one update epoch, and a common held-out set of 24 arithmetic prompts *within each seed*. The full-feedback arm purchases six labels per group; uniform and joint EWMA purchase three. Every arm performs 60 optimizer steps per seed. The frozen logs report nonzero-gradient steps and nonzero adapter changes, and their replay audits pass, though GPU backward is not independently rerun by the journal audit.

Uniform and joint EWMA are identical on every *final* paired held-out item (Table 3). Their seed-paired final accuracy difference is exactly zero in this sample, whereas both are 0.01389 below the six-label full arm in mean seed accuracy. With only three seed clusters, a zero-width sample contrast cannot certify population equivalence. The archived descriptive t interval for joint minus full is $[-0.07365, 0.04587]$. The two budgeted policies make different acquisition decisions after calibration; their identical final accuracy does not mean that their parameter trajectories coincide.

The normalized trapezoidal area under the observed held-out learning curve is 0.81713 for full, 0.79977 for uniform, and 0.79282 for joint EWMA, averaged over seeds. Joint minus uniform is -0.00694, with a descriptive seed-cluster interval $[-0.04145, 0.02756]$. Among seeds that reach 95% held-out accuracy by a scheduled evaluation, the first recorded crossing is at mean step 40 for full, 50 for uniform, and 55 for joint EWMA; the third seed is right-censored at step 60 for all three arms. These crossing summaries do not identify the exact step of improvement between evaluations.

An exploratory, post-hoc fixed stress bank evaluates saved checkpoints on 96 other prompts per seed, without changing the training. The final accuracies by stratum are given in Table 4; three-term addition remains especially difficult. At the final checkpoints, the uniform policy alone answers two of the 288 stress items correctly, while joint EWMA alone answers none. These counts share task panels across arms and do not warrant conventional independent-item significance tests.

Pairwise LoRA adapter distances confirm distinct trained parameters. Across seeds, the final raw adapter ℓ_2 distances between uniform and joint EWMA are 0.08441, 0.05611, and 0.06023; the reconstructed effective-update distances are 0.03999, 0.03513, and 0.04046. Parameter divergence does not imply an accuracy advantage. The acquisition design used during training relies on projected token scores. The following diagnostic tests how the resulting risk criterion transfers to actual LoRA gradients.

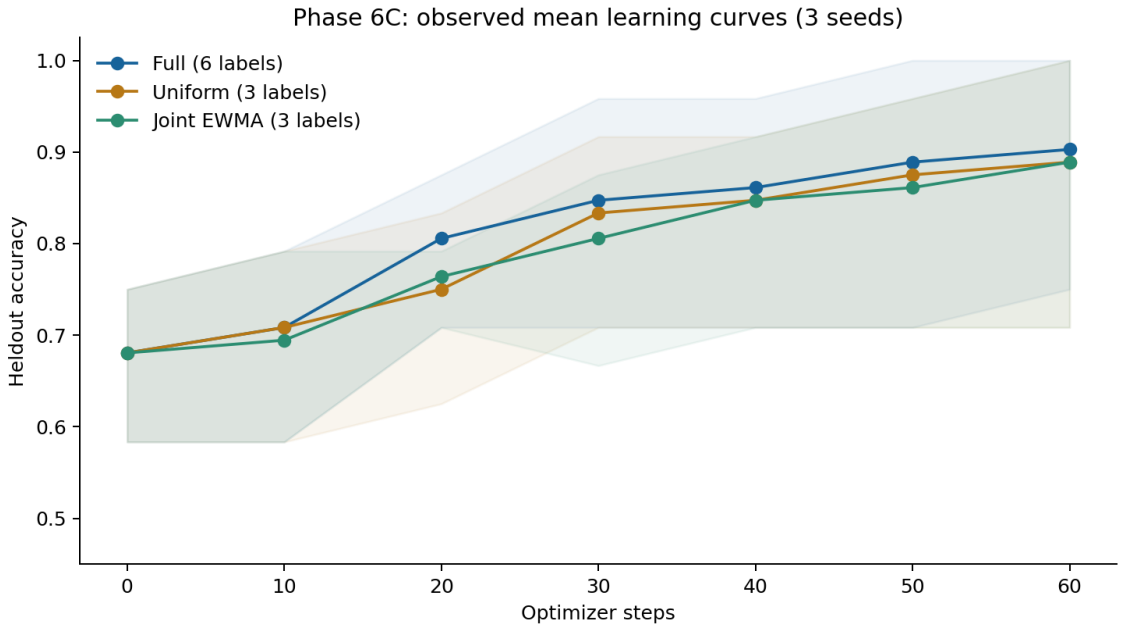


Figure 3: Mean held-out accuracy at recorded optimizer checkpoints. Shading denotes the observed *minimum-to-maximum seed range*, not a confidence interval. The full arm uses twice the logical label budget of either other arm.

Table 4: Final stress-bank accuracy, mean over three seed panels. These are descriptive post-hoc evaluations, not a second confirmatory test.

Unseen stress stratum	Full	Uniform	Joint EWMA
Two-digit carry, shifted input distribution	1.0000	1.0000	0.9792
Three-digit, double carry	0.8229	0.8229	0.8229
Three-term addition	0.0833	0.0625	0.0625

6.4 Frozen-checkpoint LoRA-gradient risk (Phase 6D)

Using the completed Phase 6C runs, we froze the uniform-trained and joint-EWMA-trained adapters at steps 0, 30, and 60 for each of the same three seeds. We evaluated four previously unused prompts from each of two task families: two-digit addition and three-digit double-carry addition. This gives $3 \times 2 \times 3 \times 2 \times 4 = 144$ group conditions. Each condition uses one fixed set of $G = 6$ model completions, their six gold rewards, and six actual autograd gradients with respect to trainable LoRA parameters; no optimizer step is taken. Both acquisition designs face those same observations, have $B = 3$ and $\pi_i = 1/2$, and are evaluated by enumerating all $\binom{6}{3} = 20$ subsets. The joint design is proposed from historical observations before any current reward is revealed. The task bank and source-checkpoint hashes were locked before evaluation. This is an exploratory mechanism check, not another training trial.

For completion i, we differentiate its token log-probability sum divided by the fixed generation length $T = 64$. Let h_i denote that parameter gradient and $d_i = (h_i - \bar{h})/(G-1)$. With observed binary reward r_i and subset indicators Z_i, the full and sampled linear gradients are $g_{\text{full}} = \sum_i d_i r_i$ and $\widehat{g} = \sum_i d_i Z_i r_i / \pi_i$. We compute

$$\mathcal{R}_{\text{LoRA}}(Q) = \sum_{|T'|=3} q_{T'} \left\| \sum_i d_i r_i \left(\frac{\mathbf{1}\{i \in T'\}}{\pi_i} - 1 \right) \right\|_2^2 . \tag{12}$$

This is the exact conditional risk of a first-order on-policy LOO–HT gradient, evaluated from the real 6×6 gradient Gram matrix. The sketch risk uses the same rewards and subset laws with the original three-dimensional scores in place of h_i. The training implementation corresponds to the zero reward-prediction baseline $m = 0$ of (4); neither risk measure includes AdamW preconditioning or later optimization dynamics.

Table 5: Phase 6D conditional query-risk reductions, Uniform minus Joint EWMA divided by summed Uniform risk. Positive values favor Joint EWMA. Comparisons are paired within each frozen checkpoint and completion group; the two source arms are distinct trained trajectories. Percentages describe these archived groups, not an independent-sample confidence statement.

Frozen-condition subset	Groups	Sketch reduction	Real LoRA reduction
Step 0: identical proposals	48	0.00%	0.00%
Step 30	48	31.16%	−6.83%
Step 60	48	23.01%	21.31%
Steps 30–60, pooled	96	26.91%	3.27%
Uniform-source checkpoints	48	29.04%	20.94%
Joint-source checkpoints	48	24.25%	−11.09%

At step 0 the designs coincide, providing an exact negative control. At later checkpoints all 96 proposals differ at the pair-inclusion level, but true-risk changes are not uniformly favorable (Table 5). Across the three seeds the pooled true-risk reductions are 16.80%, −22.46%, and 10.70%, respectively. Averaging these three seed-specific percentages gives 1.68%, with a descriptive seed-cluster t interval of approximately $[-50.81\%, 54.16\%]$. The 3.27% pooled reduction in the table instead weights groups by their baseline risk; neither summary establishes a stable positive population effect.

Using a numerical tie rule $|\mathcal{R}_U - \mathcal{R}_J| \le 10^{-8} \max\{\mathcal{R}_U, 10^{-12}\}$, the 96 later groups contain 27 improvements, nine deteriorations, and 60 ties in real-gradient risk. The sketch agrees in direction in 34 of the 36 groups with a non-negligible true-risk change, but substantially overstates the aggregated improvement. Many late completions yield identical or near-identical rewards and gradient geometry,

leaving little scope for coupling gains. The diagnostic therefore rules out treating a sketch-risk reduction as a quantitative certificate for the true LoRA-gradient risk. It does not measure a reduction in downstream optimizer error or contradict the held-out accuracy tie in Table 3.

7 Interpretation and limitations

Design-unbiasedness and the covariance decomposition concern a linear LOO estimator with known inclusion probabilities. The synthetic experiments assess conditional acquisition risk using projected policy scores. Their comparisons depend on residual structure and the accuracy of historical moment estimates; joint coupling is not uniformly better than diagonal-scaled balancing.

The LoRA training experiment measures held-out predictions after nonlinear optimizer updates. Uniform and joint EWMA select different rewards and produce distinct adapters but agree on every final held-out prediction across the three seed panels. This observed agreement does not establish population equivalence. The subsequent frozen-checkpoint diagnostic uses genuine LoRA autograd, closing the earlier measurement gap between projected scores and actual first-order gradients at the inspected checkpoints. Its mixed results show that sketch-risk improvements need not transfer in magnitude or even sign. It does not identify a unique cause for the unchanged final accuracy: optimizer preconditioning, finite training duration, task difficulty, and the coarse binary accuracy measure remain possible explanations, not independently verified mechanisms.

Fixed-size coupling has a classical predecessor in auxiliary balancing; the archive's cube-style procedures should not be read as implementations of every landing step in the original cube method. The exact arithmetic oracle incurs a logical label budget but negligible verifier expense. Synthetic parameters were explored during development, the arithmetic task family was selected after a pilot, and the stress bank was introduced post hoc. Phase 6D reuses those three seed trajectories, so its 144 groups are not independent training replications; the intervals are descriptive. These results do not establish performance for human preferences, broader language-model tasks, or standard-deviation-normalized, multi-epoch clipped GRPO. The model revision was recorded as `main`. The lightweight archive contains the frozen Phase 6D gradient Gram records but omits the original optimizer checkpoints; the CPU replay verifies saved computations, not independent GPU autograd. Exact model-level replay may depend on the model revision, environment, and saved adapters.

8 Conclusion

Model-assisted selective reward observation preserves the unnormalized LOO gradient in design expectation. Correlated prediction residuals can affect its additional query risk when selection decisions are coupled, with the effect determined by policy-score geometry. For small groups, the optimal fixed-budget coupling is tractable but reduces to classical auxiliary balancing. Historical calibration helps in some synthetic regimes and hurts in others. Actual LoRA gradients confirm that sketch-risk reductions are not reliable quantitative proxies for parameter-gradient benefits. The completed training study finds distinct acquired rewards and learned adapters, but no final held-out accuracy improvement over uniform acquisition.

Data and code availability

The companion archive contains source, configurations, tests, machine-readable results, frozen Phase 6D per-group Gram records, and plotting inputs. The repository URL is:

`https://github.com/SequentialDecisionResearch/correlation-aware-selective-reward`

Reported tables and figures can be regenerated or verified from archived data without GPU training, using the documented CPU audit scripts. The underlying autograd and model updates cannot be independently recomputed from this lightweight bundle because it omits original optimizer checkpoints and model weights.

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A Proofs and sampling details

Proof of Theorem 1. For $i \neq j$, mean separability and conditional independence of A_i and A_j imply $\mathbb{E}[s_i R_j \mid X] = \mathbb{E}[s_i \mu(X, A_j) \mid X] = 0$, since $\mathbb{E}[s_i \mid X] = 0$. For the diagonal term the score identity gives $\mathbb{E}[s_i R_i \mid X] = \nabla_\theta \mathbb{E}_{A \sim \pi_\theta(\cdot|X)} \mu(X, A)$. In the LOO sum, each R_i has coefficient $+1$ in its own advantage, while its appearances in other advantages multiply scores independent of its conditional mean. Averaging over i and X proves the assertion. □

Proof of Theorem 2. Condition on $(R, \mathcal{F}_0)$. Then m_i, π_i, R_i are fixed and $\mathbb{E}_Z[Z_i/\pi_i] = 1$, proving (5). Put $e_i = R_i - m_i$ and $U_i = Z_i/\pi_i - 1$. Since $\widehat{R}_i - R_i = U_i e_i$ and U is conditionally independent of R, $\mathrm{Cov}_Z(\widehat{R}_i, \widehat{R}_j \mid R, \mathcal{F}_0) = (C_Q)_{ij} e_i e_j$. The law of total covariance gives

$$\mathrm{Cov}(\widehat{R} \mid \mathcal{F}_0) = \mathrm{Cov}(R \mid \mathcal{F}_0) + \mathbb{E}[(C_Q \odot ee^\top) \mid \mathcal{F}_0].$$

Both M and C_Q are $\mathcal{F}_0$-measurable, which yields (6); taking the trace of $M(C_Q \odot V)M^\top$ proves (7). □

Proof of Proposition 3. For $i \neq j$, independence gives $\pi_{ij} = \pi_i \pi_j$; for $i = j$, $\pi_{ii} = \pi_i$. Substitution into (7) gives (8). Write $a_i = V_{ii}\|M_{\cdot i}\|^2$. The interior Lagrangian first-order condition for minimizing $\sum_i a_i/\pi_i$ subject to $\sum_i c_i \pi_i = C$ is $-a_i/\pi_i^2 + \lambda c_i = 0$. Truncating the resulting $\sqrt{a_i/(\lambda c_i)}$ at a positive floor and one gives the constrained rule; if $a_i = 0$, optima need not be unique. □

Proof of Proposition 4. If $B = 1$, only singleton sets occur and $q_{\{i\}} = \pi_i$. If $B = G - 1$, each subset omits exactly one index, whose omission probability is $1 - \pi_i$. For additive pair coefficients, $\sum_{i<j}(u_i + u_j)\pi_{ij} = \sum_i u_i \mathbb{E}[Z_i \sum_{j \neq i} Z_j] = (B-1)\sum_i u_i \pi_i$, which is fixed. For (ii), choose a Gram factorization of PSD V. With $K_{ij} = M_{\cdot i}^\top M_{\cdot j}$, $\langle w_i, w_j \rangle = V_{ij} K_{ij}$. Expanding the squared norm on the right of (10) and taking the query expectation yields (7). □

Proof of the plug-in bound (11). Write $F(Q, V) = \langle W_Q, V \rangle$. Since $F(\widehat{Q}, \widehat{V}) \leq F(Q^\star, \widehat{V})$,

$$\begin{aligned} F(\widehat{Q}, V) - F(Q^\star, V) &\leq \langle W_{\widehat{Q}}, V - \widehat{V} \rangle + \langle W_{Q^\star}, \widehat{V} - V \rangle \\ &\leq (\|W_{\widehat{Q}}\|_{\mathrm{F}} + \|W_{Q^\star}\|_{\mathrm{F}}) \|\widehat{V} - V\|_{\mathrm{F}}. \end{aligned}$$

Optimality of $Q^\star$ gives the lower bound of zero. □

Exact-size uniform control. For uniform sampling of B among G responses, $\pi_i = B/G$ and $\pi_{ij} = B(B-1)/(G(G-1))$ for $i \neq j$. Consequently $(C_Q)_{ij} = -(G-B)/(B(G-1))$ off the diagonal. This serves as a simple inclusion-probability unit test.

B Archived evidence and reproducibility map

Paths below are relative to the separate reviewer reproducibility archive.

Controlled designs (Table 1, Figure 1).
`results/phase2_full/summary.json`, `results/phase2_full/all_trials.csv`, and `results/phase2_figures/`.

Independent synthetic holdout.
`results/phase4_frozen_holdout_full/holdout_summary.json`.

Six-world drift comparison (Table 2, Figure 2).
`results/phase5_full/phase5_summary.json` and `results/phase5_figures/`.

Task-family calibration.
`results/phase6b_calibration2_v1/calibration2_summary.json`.

LoRA final accuracy (Table 3).
`results/phase6c_cal2_seed6207/phase6c_summary.json` (also seeds 6219 and 6251) and `results/phase6c_cal2_multiseed/`.

Learning curves and stress test.
`results/phase6c_v15_diagnostics/phase6c_learning_curve_stats.json` and `results/phase6c_v15_diagnostics/phase6c_diagnostic_summary.json`.

Adapter distances and item-level predictions.
`results/phase6c_v15_diagnostics/phase6c_adapter_distances.json` and `results/phase6c_v15_diagnostics/phase6c_item_predictions.csv`.

Frozen real-gradient risk (Table 5).
`results/phase6d_gradient_v1/phase6d_manifest.json`, `results/phase6d_gradient_v1/phase6d_task_bank.json`, `results/phase6d_gradient_v1/phase6d_summary.json`, `results/phase6d_gradient_v1/phase6d_group_risk_comparison.csv`, and `results/phase6d_gradient_v1/groups/` (144 SHA-256-checked records). CPU-only replay: `scripts/reproduce_phase6d_archived.py`.

The precomputed figures in this LaTeX package are copied from the archived plotting outputs; their inputs are included in the separate reviewer reproducibility package. Phase 6C numerical outputs were audited by journal/reward/design replay and saved-adapter comparisons. Phase 6D adds a reproducible arithmetic replay of all 144 conditional risks from the saved true-gradient Gram matrices, not an independent CUDA backward pass. Inference uses independent random seeds as descriptive clusters, rather than treating all checkpoints or task questions as independent experiments.